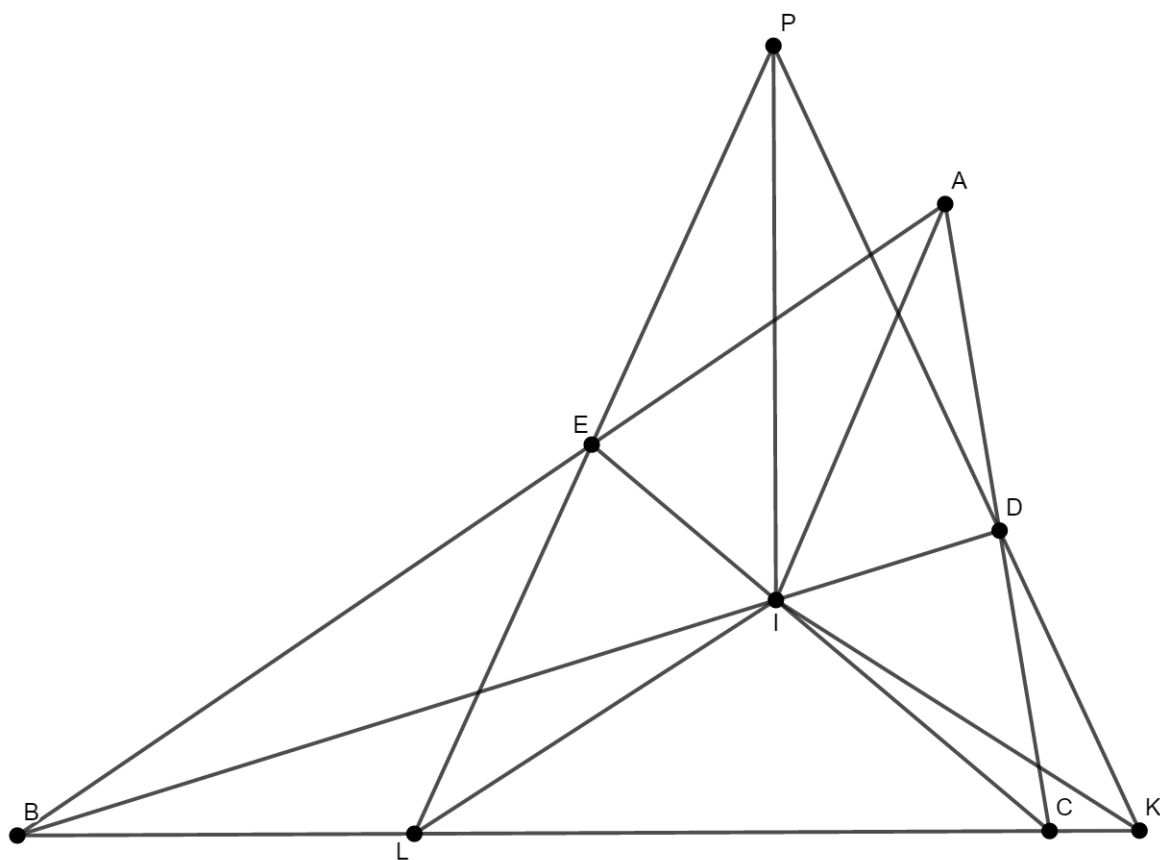


### 4.3.31



Let  $K$  and  $L$  be reflections of  $A$  in  $BD$  and  $CE$ . Then  $K$  and  $L$  lies on  $BC$ .  
 We have  $\angle PKL = \angle BAC = \angle PLK$ ,  $PK = PL$ . Also,  $IK = IA = IL$ , so  $PI \perp KL$ .